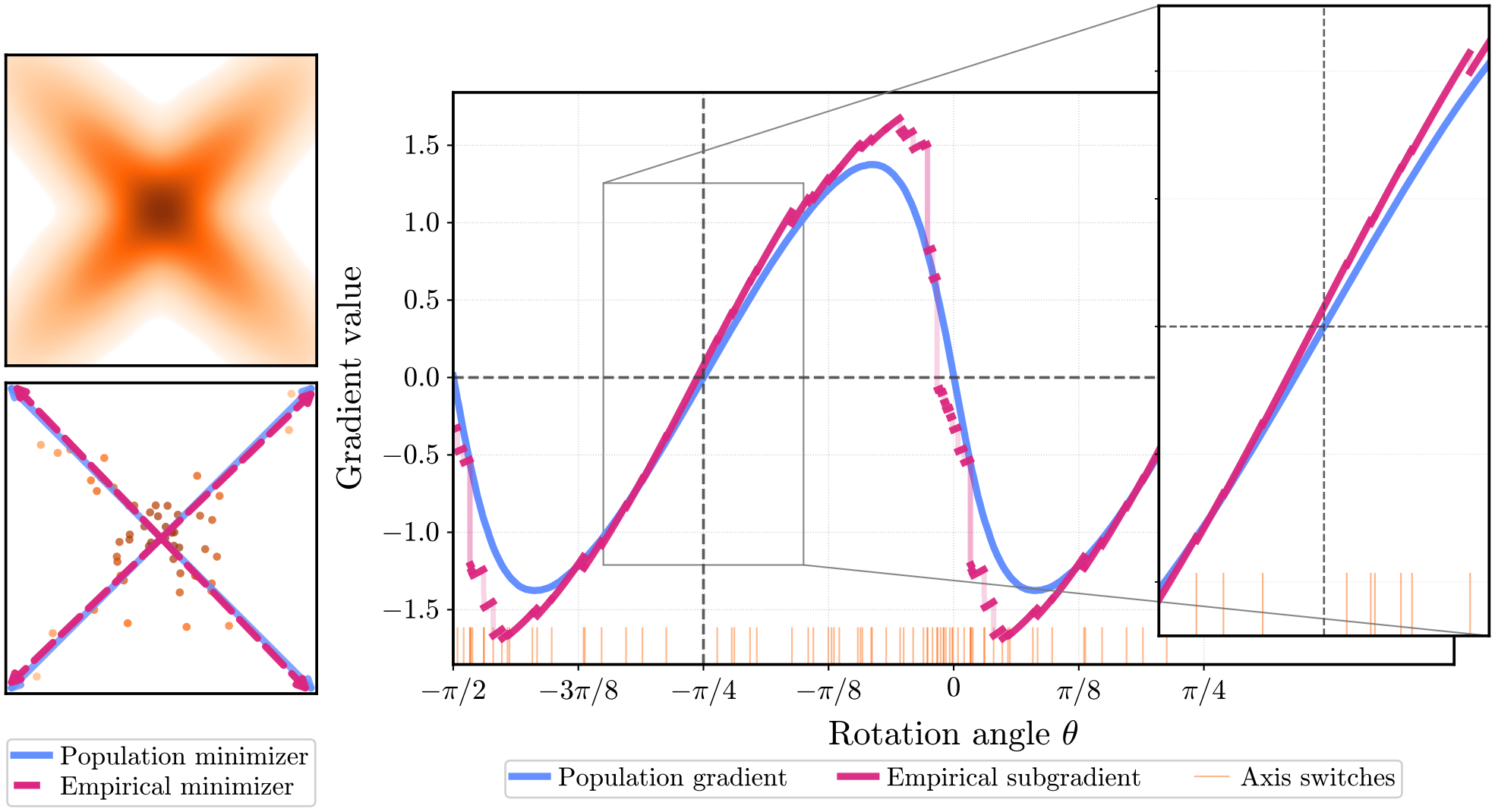

Gradient value
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−π/2
−3π/8
−π/4
−π/8
0
π/8
π/4
Rotation angle θ
Population gradient
Empirical subgradient
Axis switches
Population minimizer
Empirical minimizer